System Configuration & Service Administration

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Audience: Platform administrators • Infrastructure engineers • Technical operations teams

Purpose and Scope

This guide describes role-based access, configuration inventory, node onboarding, service lifecycle controls, configuration validation, observability, and recovery verification. It emphasizes safe administrative practices: approval, prerequisite checks, change records, post-change validation, and rollback planning.

Quick Reference

Task	Navigation/reference
Review infrastructure	Administration > Infrastructure > Overview
Inspect node health	Infrastructure > Nodes > Node Details
Review managed services	Administration > Services > Inventory
Review change history	Administration > Audit > Change History
Review background tasks	Administration > Tasks > Activity

1. Getting Started

1.1 Access Requirements

Access is restricted to authenticated users assigned a system-administration role. Available actions depend on an assigned permission set and the environment selected.

Role	Typical access	Restrictions
Viewer	Read inventory, health, logs	Cannot modify settings
Operator	View resources; run approved maintenance actions	Cannot change policy or security settings
Administrator	Edit local configurations; administer service lifecycles	Requires change approval for high-impact operations

Note: Role names and permissions in this sample are illustrative. Follow the organization’s access-control policy for actual privileges.

1.2 Sign in and Select an Environment

1. Sign in to the Northstar Operations Platform using an authorized account.
2. Open Administration > System Configuration.
3. Confirm the selected environment in the upper-right Environment selector.
4. Verify the environment banner displays the expected mode (for example, LAB or PRODUCTION).
5. Open Infrastructure > Overview to review current resource health before making changes.

Caution: Do not proceed with a configuration change when the target environment cannot be positively identified.

1.3 Navigation and Status Indicators

Indicator	Meaning	Administrator action
Healthy	All monitored checks pass	No action required
Warning	At least one check is degraded	Review details and assess impact
Critical	A service or dependency is unavailable	Escalate under incident procedure
In progress	A background task is executing	Avoid conflicting changes
Failed	Task ended unsuccessfully	Inspect task log before retry

2. Infrastructure Inventory

2.1 Review Sites and Nodes

The Infrastructure Overview lists sites and their managed nodes. Each row displays site name, environment, connectivity status, service health, last check-in, and node count.

1. Navigate to Administration > Infrastructure > Overview.
2. Select a site to open Site Details.
3. Expand Nodes to inspect node roles, hostnames, software versions, and monitoring state.
4. Select a node to open Node Details and review hardware capacity, endpoints, and assigned services.

Note: The system presents remote sites as read-only in this illustrative workflow. Only the currently selected local site can be modified.

3. Node Registration and Configuration

3.1 Pre-change Checklist

- Approved change ticket identifies target environment and maintenance window.
- Hostname, network assignment, and node role are confirmed against the approved design.
- Latest configuration export and tested recovery approach are available.
- Monitoring and dependencies are healthy or a deviation is documented.
- An authorized peer or reviewer is assigned for high-impact changes.

3.2 Register a New node

1. From Infrastructure > Nodes, select Add Node.
2. Enter the approved node name and management address.
3. Choose its assigned role (for example, application or data processing).
4. Assign the node to a site and configuration profile.
5. Select Validate to check required fields and inventory conflicts.

6. Review the validation result, then select Save Draft.
7. After approval, select Register Node and monitor the task in Administration > Tasks > Activity.
8. Confirm the node appears in inventory and passes connectivity and heartbeat checks.

3.3 Example Configuration Manifest

The following JSON illustrates the structure of a proposed node registration payload. It is not a deployable configuration file.

```
{
  "environment": "lab",
  "site": "demo-east",
  "node": {
    "name": "lab-app-02",
    "managementAddress":
"192.0.2.24",
    "role": "application",
    "profile": "baseline-v1"
  },
  "changeReference": "DEMO-1042"
}
```

Note: 192.0.2.0/24 is reserved for documentation examples. Never place passwords, access tokens, or private keys in a sample configuration.

3.4 Edit a Node

1. Open Infrastructure > Nodes and select the target node.
2. Select Edit Configuration and compare current values with the approved change.
3. Modify only the fields identified in the change record.
4. Select Validate and review warnings, dependencies, and predicted impact.
5. Save the draft, request approval as required and apply during the approved window.
6. Confirm heartbeat, services, and monitoring return to their expected states.

Warning: Changing a hostname or advertised management address may break control-plane connectivity. Treat these fields as high impact.

4. Managed Services

4.1 View the Service Inventory

The Services Inventory provides installed version, enabled version, instance count, allocation health, dependency status, and last deployment time for each managed service.

1. Navigate to Administration > Services > Inventory.
2. Filter by site, status, or service name.
3. Select a service to open Service Details.
4. Review the Overview, Packages, Instances, Logs, and Deployment History tabs.

4.2 Service Lifecycle Actions

Action	Purpose	Validation after completion
Enable	Activate a deployed, approved package	Healthy instances and dependency checks
Disable	Stop serving traffic for the selected service	Dependent services reviewed and routing updated
Restart	Recycle instances to resolve a known condition	Instances return healthy; error rate stabilizes
Upgrade	Activate an approved newer package	Version matches target; checks pass
Rollback	Restore a prior approved package	Previous known-good version active

4.3 Upgrade a Service Package

1. Open Services > Inventory and select the target service.
2. Confirm current version, package provenance, dependencies, and health metrics.
3. Review the approved deployment plan and rollback criteria.
4. Select Packages and choose the approved target version.
5. Select Validate Deployment; resolve any blocking dependency errors.
6. Start the deployment in the authorized maintenance window.
7. Monitor Deployment History and the Tasks Activity view until the task completes.
8. Verify running version, instance health, logs, and operational smoke tests.
9. Record results and close the change only after the acceptance criteria pass.

Note: Never infer successful deployment solely from a green task status. Validate application behavior and downstream integrations.

4.4 Rollback Decision Criteria

A rollback should be initiated under the approved plan when a required health check fails, error rates exceed the agreed threshold, a critical dependency is unavailable, or a functional acceptance test fails. Record the triggering condition and restore the last known-good package according to the environment-specific runbook.

5. Configuration Policies and Field Management

5.1 Configuration Profiles

Configuration profiles group validated settings that may be applied to one or more nodes or services. Each profile includes an identifier, revision, owner, approval reference, and effective date.

1. Navigate to Administration > Configuration > Profiles.
2. Select a profile and review assigned resources and effective values.
3. Choose Create Revision rather than editing an approved baseline in place.

4. Modify documented parameters, enter a change rationale, and run validation.
5. Submit the revision for review and deployment approval.

5.2 Field-level Options

Property	Definition	Example behavior
Required	Prevents submission without a value	Site code must be provided
Read-only	Displays information that cannot be edited	Generated resource ID
Default	Supplies an initial value	Audit logging enabled
Conditional	Applies a value only when conditions match	Site-specific feature flag

5.3 Configuration Precedence

For this illustrative platform, settings are resolved in this order: (1) approved resource override, (2) site-level profile, (3) organization baseline. The effective value and its source are visible in the configuration viewer. This precedence is an example and must not be assumed for other platforms.

5.4 Configuration validation

Before applying a revision, confirm schema validity, required properties, reference integrity, dependency compatibility, and policy conformance. Review the change comparison and capture the validation outcome in the associated ticket.

6. Health, Monitoring, and Task Status

6.1 Storage and Resource Utilization

The Health panel displays storage utilization, CPU and memory trends, service availability, and recent alerts. Thresholds below are illustrative monitoring-policy examples rather than production operating limits.

Metric	Example warning	Example critical
Available disk space	Below 20%	Below 10%
Service health check	Intermittent failure	Sustained failure
Task execution	Exceeds expected duration	Fails or times out

6.2 Review Task Details

1. Open Administration > Tasks > Activity.
2. Filter by change reference, task type, site, or status.
3. Select the task to view timestamps, execution stages, and validation messages.
4. Export the task summary or link it to the change record.

7. Troubleshooting and Recovery Verification

7.1 Node does not report healthy

Check	Expected result	If unsuccessful
Environment selection	Correct site and environment	Select correct scope; do not modify other sites
Heartbeat	Recent successful check-in	Review network reachability and monitoring alerts
Profile validation	No blocking configuration errors	Correct draft or restore approved baseline
Assigned services	Required services running	Inspect service details and dependency logs

7.2 Service Upgrade Fails Validation

1. Open the failed task and record its error code and affected component.
2. Verify package version, configuration schema, and declared dependencies.
3. Review service logs for the relevant time window; avoid collecting secrets.
4. Do not retry blindly. Resolve the underlying condition or escalate to the service owner.
5. If the deployment changed live state, follow the approved rollback plan.
6. Repeat health and functional verification before closing the incident.

7.3 Post-change Verification Checklist

- The intended environment and target resources match the change record.
- The active package/profile matches the approved version.
- Required nodes and instances are healthy.
- Operational smoke tests pass, including affected integration paths.
- No unresolved high-severity alerts or unexplained task failures remain.
- Task logs, validation evidence, and rollback status are recorded.

8. Auditability and Configuration Governance

Administrators should maintain an auditable record of each material change, including requestor, approver, scope, version before/after, deployment time, verification evidence, and rollback outcome. Use a centralized change-management system and approved retention policy. Exclude passwords and secret values from exported records.

Appendix A. Example Change Record

Field	Illustrative entry
Change reference	DEMO-1042
Purpose	Register second lab application node
Environment	LAB
Risk classification	Moderate; no production traffic
Prerequisites	Approved baseline, profile validation, available capacity
Implementation	Create node record; validate; register; monitor
Verification	Connectivity, heartbeat, assigned services, smoke test
Rollback	Remove draft/registration using approved recovery runbook
Evidence	Task summary, validation result, reviewer sign-off